Modelling of Orthophosphate and Zinc Inputs to Rivers from Diffuse and Point Sources using a Bayesian Load Apportionment Model with Time-Varying Parameters

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Executive Summary

Effective management of river water quality in England depends on whether pollution originates from point sources, such as sewage treatment work (STW) effluents, or diffuse sources, such as agricultural runoff, as each requires fundamentally different regulatory responses. Under the Water Framework Directive, many English river bodies continue to fail Good Ecological Status, yet the balance between point and diffuse sources has shifted considerably since 2000 as point-source regulation has tightened. Understanding where this balance now stands, and how it has evolved, is essential for targeting future investment effectively.

This report investigates point and diffuse-source contributions to orthophosphate and dissolved zinc concentrations at two contrasting sites: the Thames at Teddington Weir, a large urbanised catchment with a well-documented history of STW regulation, and the Wyre at St Michaels, a smaller predominantly agricultural catchment, over the 2000–2024 period. The Load Apportionment Model was applied in a Bayesian framework to quantify uncertainty in source apportionment, and extended to allow point-source contributions to change over time rather than assuming them fixed, better reflecting the evolving regulatory and catchment conditions of a 25-year monitoring record.

For zinc, persistent point-source dominance was found at the Thames and diffuse-source dominance at the Wyre throughout the period. However, the monitoring data for zinc do not show the clear contrast between point and diffuse behaviour that the framework relies upon, and uncertainty in these apportionments is too wide to reliably guide compliance decisions at either site. Further methodological development would be needed before zinc source apportionment can inform targeted Environment Agency (EA) intervention.

The time-varying framework consistently outperformed the traditional fixed-parameter approach at sites where catchment conditions changed most rapidly, and is recommended as the preferred tool for long-term source apportionment analysis at regulated catchments. Extending this approach to all 135 Harmonised Monitoring Scheme sites across England would allow the EA to systematically identify catchments where point-source infrastructure investment would yield the greatest water quality improvement, versus those where diffuse source management has become the binding constraint on achieving Good Ecological Status.

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1 Introduction

Effective management of river water quality in England requires the ability to distinguish between point-source pollutants, such as sewage treatment work (STW) effluents and industrial discharges, and diffuse-source pollutants, such as agricultural runoff and road drainage. Under the EU Water Framework Directive (WFD; 2000/60/EC), river bodies across England are required to achieve Good Ecological Status (GES), yet many continue to fail these standards [10, 15]. Identifying the dominant pollution source is therefore essential for targeting mitigation efforts effectively.

Excess orthophosphate promotes harmful algal growth and eutrophication, causing dissolved oxygen fluctuations that disrupt aquatic ecosystems [6]. Elevated concentrations of dissolved metals such as zinc are directly toxic to aquatic organisms and pose risks to human health [9]. As point-source contributions have declined following regulatory intervention, notably mandatory phosphorus stripping at larger STWs under the Urban Wastewater Treatment Directive, diffuse sources are increasingly the binding constraint on achieving Good Ecological Status. Quantifying the relative contributions of point and diffuse sources over time has therefore become a central water management challenge.

The Load Apportionment Model (LAM), introduced by [2], exploits the contrasting hydrological signatures of these two source types, expressing total pollutant concentration as the sum of two power-law functions of river flow. Requiring only routine concentration and flow monitoring data, the LAM offers a parsimonious and interpretable framework for source apportionment. However, its assumption of fixed parameters is inappropriate over long time series where catchment conditions evolve, a limitation motivating this study. We therefore propose a Bayesian implementation of the LAM to formally quantify uncertainty in source apportionment estimates, and a time-varying extension (TV-LAM) to accommodate structural non-stationarity in the flow-concentration relationship.

This report applies both frameworks to orthophosphate and zinc at contrasting sites on the rivers Thames and Wyre, comparing their ability to characterise point and diffuse source contributions over a 25-year period and evaluating whether accounting for temporal variation meaningfully affects source apportionment conclusions.

2 Data Overview and Pre-processing

2.1 Water Quality Concentration Data

Spot-sample water quality data were obtained from the Harmonised Monitoring Scheme (HMS) network, which provides a long-term record of nutrient and metal concentrations at 135 sites on England's principal rivers [13]. Although GB-wide official reporting ceased in 2013, the Environment Agency (EA) has continued sampling at the same sites under its own monitoring programme, with all samples analysed by EA laboratories and their predecessors, ensuring consistency across the full 2000–2024 study period [14].

We focus our analysis on two determinands: orthophosphate ($\mathrm{mg\,l^{-1}}$) and dissolved zinc ($\mu\mathrm{g\,l^{-1}}$), chosen to represent contrasting nutrient and metal pollutant types with differing point/diffuse source characteristics. These concentrations are investigated at two contrasting sites: *Thames at Teddington Weir*, a large, urbanised, heavily regulated catchment with a well-documented history of STW phosphorus stripping; and *Wyre at St Michaels*, a smaller, predominantly agricultural catchment expected to exhibit greater diffuse-source influence. This contrast provides a meaningful test of the LAM framework across different source regimes. Both sites were sampled at least monthly over the 2000–2024 study period, with known gaps during the 2001 foot-and-mouth outbreak and 2020–2021 COVID-19 lockdowns [14].

Observations falling below implausibly low thresholds, less than $0.005\,\mathrm{mg\,l^{-1}}$ for orthophosphate and less than $0.5\,\mu\mathrm{g\,l^{-1}}$ for dissolved zinc, were removed as likely analytical errors. This resulted in the removal of one orthophosphate observation; no zinc records were affected.

2.2 Flow Data

Daily mean river discharge (m^3s^{-1}) was obtained from the Environment Agency Hydrological Data Explorer API and matched to each water quality spot sample by date [1]. Each concentration observation was paired to the daily mean flow on the corresponding sampling date. Where multiple samples existed on the same date, these raw concentrations were averaged prior to log transformation.

River flow data included Estimated (17.8%), Suspect (9.0%), and Unchecked (0.7%) quality-flagged readings alongside Good-quality observations (72.6%). Estimated readings reflect standard EA infilling procedures and are not indicative of measurement error. Suspect readings showed no systematic association with high-flow gauge exceedance, and therefore all quality categories were retained.

2.3 Censored Observations

Table 1 summarises the proportion of censored observations at each site-determinand combination.

Table 1: Data availability summary for key sites and determinands. Censoring indicates observations recorded below the limit of detection.

Determinand	Proportion of censored data	
	Thames at Teddington Weir	Wyre at St Michaels
Orthophosphate	0.0083	0.0034
Dissolved Zinc	0.1962	0.2222

Zinc exhibits substantially more censored observations (around 20% at both sites) than orthophosphate ($< 1\%$), for which the true concentration is known only to lie below the recorded limit of detection (LoD). Censored zinc observations are concentrated in the earlier part of the study period, becoming increasingly sparse after approximately 2015, consistent with improvements in EA analytical detection capability over time, particularly from 2014 onwards

[9]. Naively substituting these values (for example, with half the LoD) or discarding them entirely introduces informative selection bias, as below-detection observations are systematically associated with lower true concentrations. Instead, each censored observation is encoded as a left-censored interval $(-\infty, \log_{10}(d_i)]$, where d_i is the recorded LoD, allowing models used in this report to integrate over this uncertainty when estimating parameters rather than treating below-detection values as known.

3 Exploratory Data Analysis

To model orthophosphate and zinc concentrations effectively, we first examine their temporal trends and relationship with river flow. Figures 1 and 2 show the concentrations of orthophosphate and zinc over time at both sites.

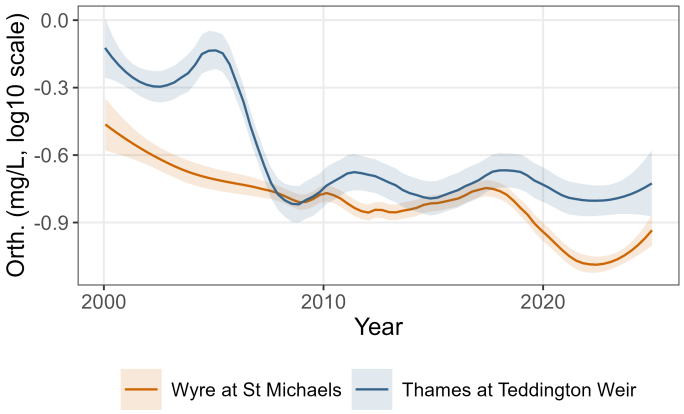


Figure 1: Log-scaled orthophosphate (mg/l) measurements over years 2000−2024 at primary sites *Thames at Teddington Weir* and *Wyre at St Michaels*.

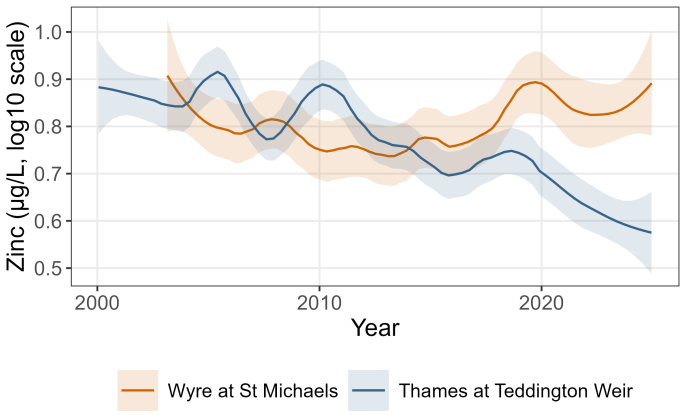


Figure 2: Log-scaled zinc ($\mu g/l$) measurements over years 2000−2024 at primary sites *Thames at Teddington Weir* and *Wyre at St Michaels*.

Orthophosphate declines clearly over the 25-year period at both sites, with the Thames exhibiting a particularly sharp reduction between 2005 and 2010 before stabilising. Zinc trends are more site-specific: the Thames shows a general decline, while the Wyre decreases until around 2012 before rising steadily to 2024. These long-run trends indicate that the LAM's assumption of fixed parameters is inappropriate over the full period, motivating the use of shorter time windows for the baseline model (Section 4), and a time-varying extension thereafter (Section 5).

Figure 3 shows the relationship between log river flow and log concentration for both determinands at both sites.

Orthophosphate exhibits a clear negative relationship with flow at both sites, consistent with dilution of a predominantly point-source signal, with diminishing returns at higher flows. Zinc, by contrast, shows a more complex and site-variable relationship: concentrations initially decrease at low flows before rising again at high flows. This upturn is consistent with zinc having a stronger diffuse-source component, whereby high-flow events that dilute point-source inputs simultaneously mobilise zinc from agricultural land and urban runoff.

The contrasting flow-concentration relationships between the two determinands motivate a modelling framework capable of separately quantifying point-source and diffuse-source contributions.

3.1 Generalised Additive Model (GAM)

Generalised additive models (GAMs) are generalised linear models that allow nonlinear functions of covariates to be incorporated in regression equations, making them well-suited for capturing nonlinear temporal trends [19]. While GAMs cannot separately quantify point-source and diffuse-source contributions, they serve here as a diagnostic tool to assess whether the flow-concentration relationship changes over time, a finding that directly motivates the time-varying extension of the LAM developed in Section 5.

We fit both an additive model,

$$\log_{10}(C_i) = \alpha + f_1(m_i) + f_2(t_i) + f_3(\log Q_i) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2), \tag{1}$$

and an interaction model,

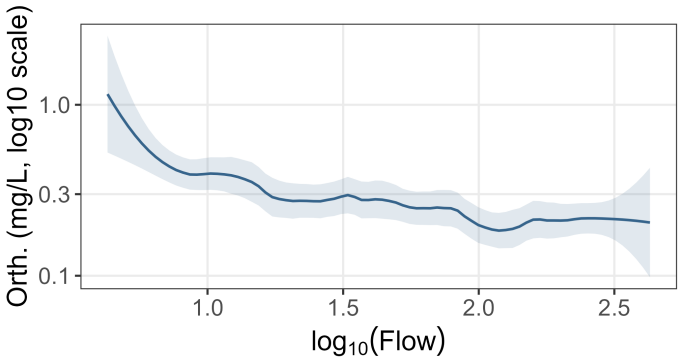
$$\begin{aligned} \log_{10}(C_i) = {} & \alpha + f_1(m_i) + f_2(t_i) \\ & + f_3(\log Q_i) + f_4(t_i, \log Q_i) + \epsilon, \\ & \epsilon \sim \mathcal{N}(0, \sigma^2), \end{aligned} \tag{2}$$

where Q_i is river discharge ($m^3 s^{-1}$), t_i is the year-date in decimal form, and m_i is a factor variable representing months of the year. Each smooth f_j is represented as a penalised regression spline, with smoothing parameters estimated by restricted maximum likelihood (REML) [19].

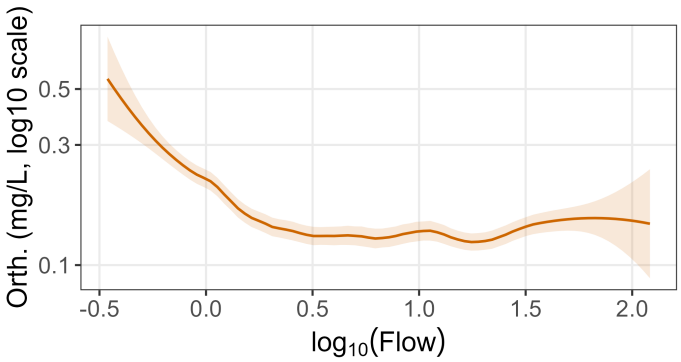
Models were constructed using the `mgcv` [19] library in `R` [17], with censored observations handled via the `cnorm()` family as described in Section 2.3. Basis dimension sufficiency was confirmed for all site-determinand combinations using `gam.check()`.

3.1.1 Model Selection and Results

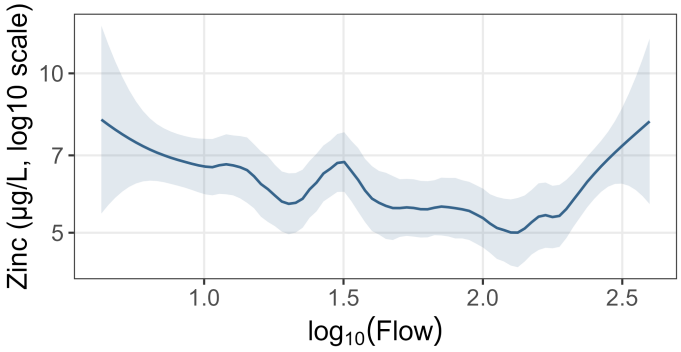
Both models were fitted for each site-determinand combination and compared via REML score (Table 2).



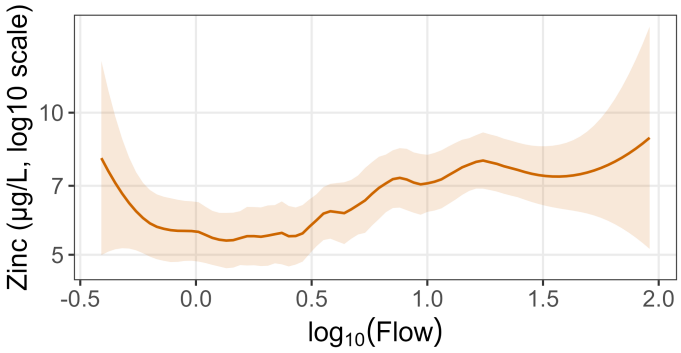
(a) Thames orthophosphate



(b) Wyre orthophosphate



(c) Thames zinc



(d) Wyre zinc

Figure 3: Log10-scaled concentration against log10-scaled river flow (m^3/s) for orthophosphate (mg/l) and zinc ($\mu g/l$) at *Thames at Teddington Weir* and *Wyre at St Michaels*.

Table 2: REML scores for additive and interaction GAM models across all site-determinand combinations. More negative values indicate better fit.

Site	Det.	Additive	Interaction
Thames	Phosphate	−74.3	−79.4
Thames	Zinc	−42.6	−44.4
Wyre	Phosphate	−179.4	−320.7
Wyre	Zinc	−3.7	−7.7

The interaction model achieved a lower REML score in all cases. The interaction term $f_4(t_i, \log Q_i)$ was statistically significant for orthophosphate at both sites (Thames: $p < 0.001$, Wyre: $p < 0.001$), with REML improvements of 5.1 and 141.3 respectively, indicating that the relationship between river flow and orthophosphate concentration changes meaningfully over the 25-year period. For zinc, the interaction term was not significant at either site (Thames: $p = 0.189$, Wyre: $p = 0.163$), with smaller REML improvements of 1.8 and 4.0. The interaction model was therefore selected for orthophosphate and the additive model for zinc.

Residual diagnostics for the orthophosphate interaction model at the Thames are shown in Figure 4.

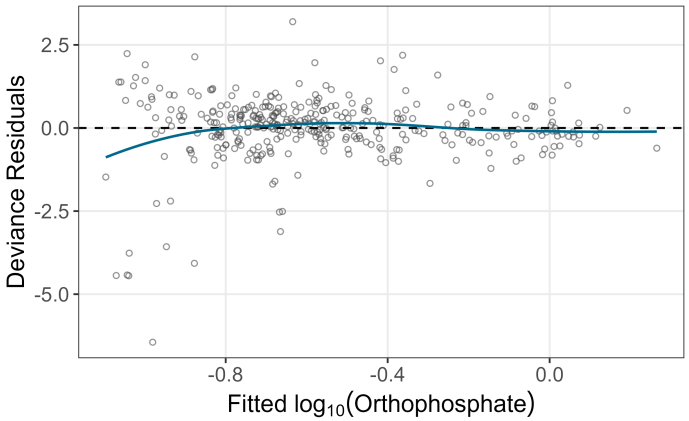


Figure 4: Residuals vs fitted values for the interaction GAM fitted to orthophosphate at the Thames, with smoothed blue LOESS trend line.

The residuals vs fitted plot shows no systematic trend or heteroskedasticity, indicating the model assumptions are adequately satisfied. Results for other site-determinand model cases were broadly similar (see Appendix A).

Table 3 summarises the estimated degrees of freedom and significance of each smooth term across all final models.

All terms were statistically significant ($p < 0.05$) except the seasonal term $f_1(m_i)$ for zinc at the Wyre (edf = 0.00, $p = 0.483$), suggesting no meaningful seasonal pattern in zinc at this site. The temporal trend $f_2(t_i)$ was highly significant and nonlinear for both orthophosphate models (edf ≈ 10.50 at the Thames and ≈ 10.25 at the Wyre), consistent with the long-term concentration decline visible in Figure 1. The interaction term $f_4(t_i, \log Q_i)$ showed notably higher complexity at the Wyre (edf = 12.65) than the Thames (edf = 2.30), suggesting the flow-concentration relationship has changed more substantially over time at the rural site.

Table 3: Estimated degrees of freedom (edf) and p-values for each smooth term across all final GAM models (additive and interaction of each site-determinand case).

		Orthophosphate Interaction model		Zinc Additive model	
Term	Site	edf	p	edf	p
$f_1(m_i)$	Thames	6.31	<0.001	4.52	<0.001
	Wyre	2.70	<0.001	0.00	0.483
$f_2(t_i)$	Thames	10.50	<0.001	1.00	<0.001
	Wyre	10.25	<0.001	3.84	<0.001
$f_3(\log Q_i)$	Thames	2.45	<0.001	3.64	<0.001
	Wyre	3.79	<0.001	4.02	<0.001
$f_4(t_i, \log Q_i)$	Thames	2.30	<0.001	—	
	Wyre	12.65	<0.001	—	

These findings establish that significant temporal non-stationarity is present in all site-determinand combinations, and the flow-concentration relationship for orthophosphate changes structurally over time. This motivates both the use of short time-windows in the baseline LAM (Section 4) and the time-varying extension developed in Section 5.

4 Bayesian LAM

The Load Apportionment Model (LAM), introduced by [2], provides the mechanistic framework to separately quantify point and diffuse-source contributions, expressing total pollutant concentration as the sum of two power-law functions of river flow.

Model. The LAM is expressed in load form, where the total pollutant load F_i (mgs^{-1}) at observation i is modelled as

$$F_i = A \cdot Q_i^B + C' \cdot Q_i^D, \tag{3}$$

where Q_i is river discharge (m^3s^{-1}), A and B are the point-source parameters, and C' and D are the diffuse-source parameters. Since concentration C_i (mgm^{-3}) equals load divided by volumetric flow, this can be expressed as

$$C_i = A \cdot Q_i^{B-1} + C' \cdot Q_i^{D-1} + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_i^2). \tag{4}$$

where the first term $A \cdot Q_i^{B-1}$ represents point-source contributions and the second term $C' \cdot Q_i^{D-1}$ represents diffuse-source contributions.

As the LAM assumes fixed parameters, it cannot accommodate long-term trends in catchment conditions. Models were therefore fitted to five-year windows: 2000–2004, 2005–2009, 2010–2014, 2015–2019, and 2020–2024, with the exception of 2000–2004 of Wyre zinc, for which monitoring commenced in 2003.

Left-censored observations are incorporated by treating the likelihood contribution as $\mathbb{P}(y_i < d_i|\theta)$, integrating over $(-\infty, \log_{10}(d_i))$, rather than substituting or excluding these values.

Priors. Weakly informative priors were used, with means informed by [2] for phosphorus and applied identically to zinc in the absence of analogous prior information (see Table 4).

Table 4: Prior specifications for models. Sign constraints are imposed as hard bounds. Parameters A and C are assigned half-normal priors as their scale depends on site-specific load magnitudes.

Parameter	Constraint	Prior $\mathcal{N}(\mu,\ \sigma)$
A	> 0	$\mathcal{N}(0,\ 10)$
B	$[0, 1]$	$\mathcal{N}(0.5,\ 5)$
C'	> 0	$\mathcal{N}(0,\ 10)$
D	> 1	$\mathcal{N}(1.5,\ 5)$

4.1 Evaluation of Model

All 19 models achieved satisfactory convergence when constructed in `brms` [5] in `R` [17], with $\hat{R} < 1.01$ and effective sample size exceeding 400 for all parameters across all site-determinand-window combinations [11]. Trace plots for these models confirmed adequate chain mixing throughout.

Residual diagnostics (Figures 5, 6 and 7) and posterior predictive checks (Figure 8) were performed on all models, with results shown in this paper for the Thames-Orthophosphate combination from 2000–2004 as an example of goodness-of-fit of our models. Results for all other baseline models can be seen in Appendix B.

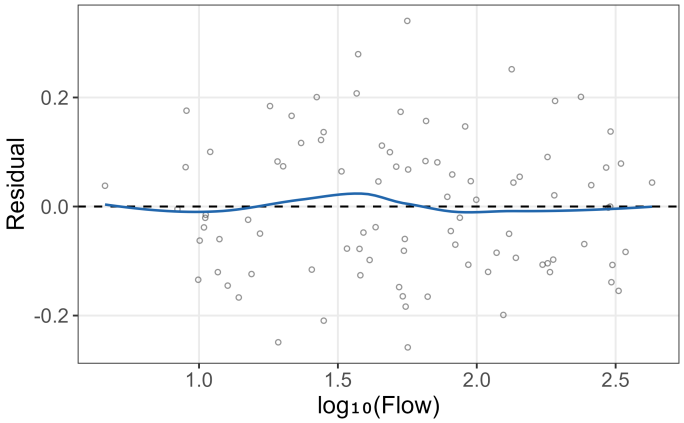


Figure 5: LAM Residual vs Flow of the Bayesian LAM on Thames-Orthophosphate data in 2000–2004, with smoothed blue LOESS trend line.

The LOESS smoothing line of the residual vs flow plot in Figure 5 is relatively flat around the zero-line, indicating the model captures the relationship between orthophosphate concentration and river flow in the Thames in 2000–2004 reasonably well. This was reflected in most site-determinand-period model combinations (see Appendix B.1), which generally exhibit similar flat LOESS behaviour.

The LOESS smoothing line of the residual vs time plot in Figure 6 however shows a systematic downward trend over time, suggesting residual temporal non-stationary is present. Even after splitting the data into shorter 5-year periods, this model systematically overpredicts later in the period and underpredicts earlier. Systematic trends such as these were shown in several models trained, indicating that a clear limitation in our LAM framework is its inability to handle temporal non-stationarity. This systematic residual trend

was present across most site-determinand combinations, as shown in Appendix B.2.

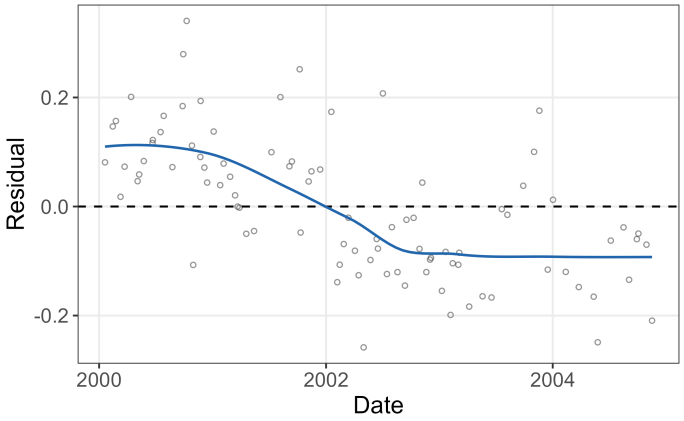


Figure 6: Residual vs Time of the Bayesian LAM on Thames-Orthophosphate data in 2000–2004, with smoothed blue LOESS trend line.

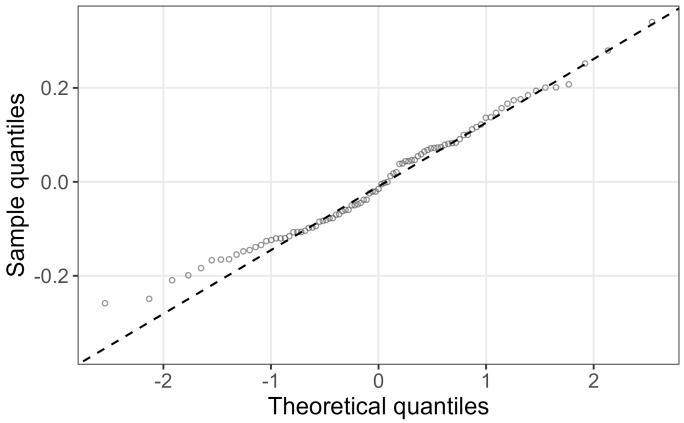


Figure 7: LAM QQ-plot of the Bayesian LAM on Thames-Orthophosphate data in 2000–2004. Dashed line = theoretical normal quantiles.

The residuals mainly follow a straight line in the QQ plot (Figure 7), suggesting that the model residuals mainly follow a normal distribution. There is however some deviation in the lower tail, suggesting a slight skewness in the lower quantiles of our concentration data. Minor lower-tail deviations were present across most models (Appendix B.3), but the normality assumption is broadly satisfied.

The posterior predictive density plot of simulated and actual $\log_{10}$ concentration here shows that the model's simulated densities envelope the observed curve reasonably well, thus indicating our model fits the data reasonably well. Posterior predictive checks for all remaining site-determinand-window combinations are shown in Appendix B.4, and demonstrate reasonably comparable fit.

4.2 Results

Following [2], the primary metric used to evaluate our model is the percentage point-source contribution to total pollutant load. This is evaluated at median river flow of

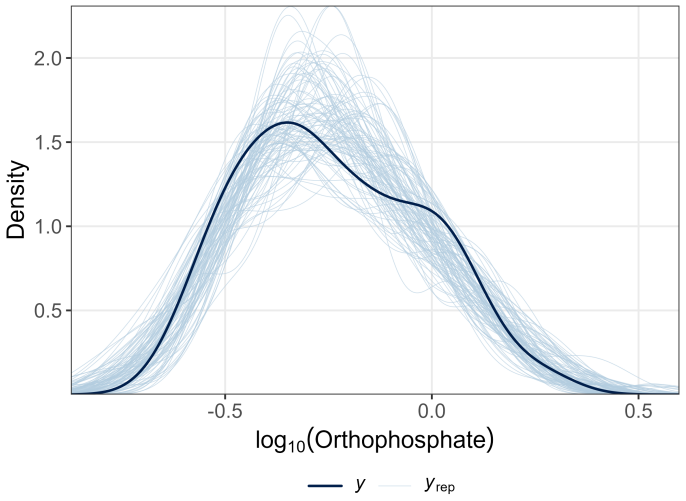


Figure 8: Posterior predictive density of simulated $\log_{10}$(Orthophosphate) concentration from Bayesian LAM at the Thames (light blue) and actual $\log_{10}$(Orthophosphate) concentration at the Thames (dark blue) in 2000–2004.

each time-period Q_{med}, and calculated as

$$\text{PS\%} = \frac{A \cdot Q_{\text{med}}^{B-1}}{A \cdot Q_{\text{med}}^{B-1} + C' \cdot Q_{\text{med}}^{D-1}} \times 100. \tag{5}$$

In the Bayesian framework, this quantity is derived from the joint posterior distribution of (A, B, C', D), yielding a full posterior distribution for PS% from which credible intervals are obtained directly.

Figures 9 and 10 present posterior median PS% estimates with 80% credible intervals across all site-determinand-window combinations (full detail of these values are shown in Appendix D).

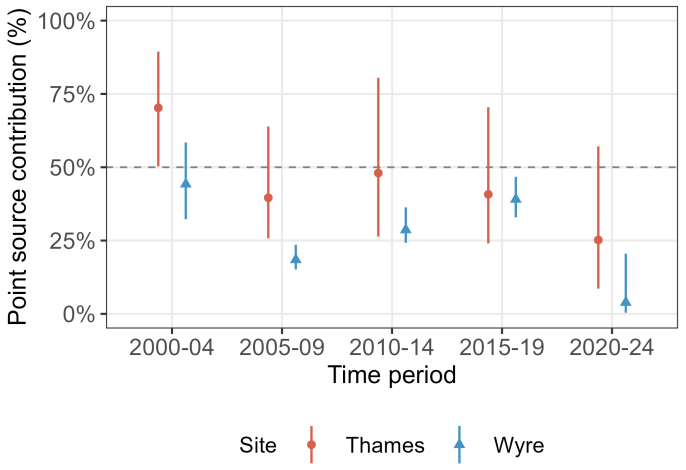


Figure 9: Estimated point-source contribution (%) of orthophosphate in the Thames and Wyre over 5-year periods from 2000 to 2024. Circle/Triangle reflects posterior median estimate, and vertical lines represent 80% of corresponding posterior median estimate.

Orthophosphate. The Thames shows clear point-source dominance in 2000–2004 (PS% = 72.8%, 80% CI: 52.7–91.2%), which decreases consistently to 25.2% (8.6–

57.1%) by 2020–2024. This trend suggests the impact of progressive STW phosphorus removal of point-source pollutants in the river Thames. The Wyre shows diffuse-source prominence across all time periods, logical given its rural, agriculturally-dominated location. Orthophosphates also show clear decline from 2000–2004 (PS% = 44.2% (32.3–58.4%) to 2020–2024 (PS% = 0.4–19.5%). Credible intervals for the Wyre are notably smaller than for the Thames across all time periods, reflected by more chemical observations being taken at this site.

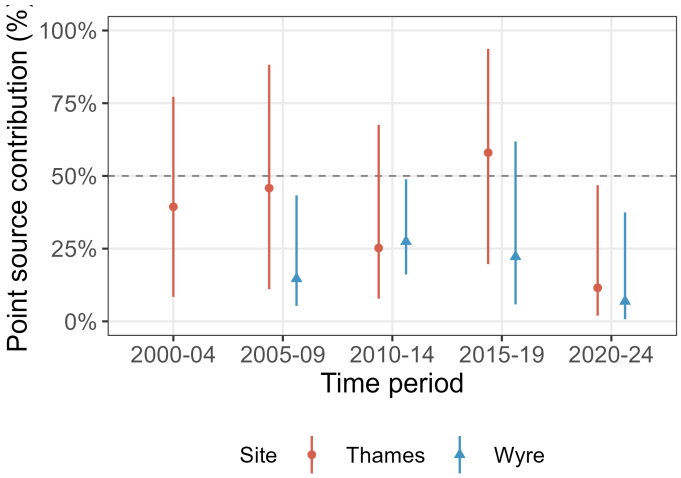


Figure 10: Estimated point-source contribution (%) of zinc in the Thames and Wyre over 5-year periods from 2000 to 2024. Circle/Triangle reflects posterior median estimate, and vertical lines represent 80% of corresponding posterior median estimate.

Zinc. The Thames does not show a clear trend over time in point-source contribution of zinc, with most time periods suggesting a contribution of around 25-75%. 2020–2024 however shows the lowest point-source dominance (PS%=11.5%, 80% CI: 4.8–25.5%). This trend is also apparent in the river Wyre, where the 2020–2024 period experienced its lowest point-source contribution of 6.9% (80% CI: 2.4–19.1%), while all other period estimates at median river flow were between 14.9–27.4%.

While these results do indeed suggest clear changes in point and diffuse-source pollution in these rivers over this 25-year period, there are clear limitations in these models that mean we should be careful before making any assumptions based on these results. Primarily, even after splitting our data into 5-year periods in an attempt to deal with temporal trends, there still exists non-stationarity in our data (as shown in Figure 6). As a result, we aim to develop a model that can handle non-stationary data.

5 Bayesian LAM with time-varying parameters (TV-LAM)

Even within five-year windows, systematic residual trends persist in the baseline LAM (Figure 6), indicating the static parameter assumption remains violated. Examining baseline posterior estimates, B and D vary across periods but with substantially overlapping credible intervals in all site-determinand cases, suggesting this variation is largely indistinguishable from noise (Appendix C). By contrast, A shows a systematic decline whose magnitude consistently exceeds posterior uncertainty, particularly at the Thames for orthophosphate, where it falls from 3.96 (95% CI: 2.45–6.66) to 0.78 (0.07–2.06). This motivates a framework in which A alone is treated as time-varying.

These results align with ecological background. A is an intercept-like scale parameter capturing the magnitude of point-source inputs, which genuinely changes over time as STW infrastructure improves and regulatory pressure evolves. B, C' and D reflect more stable catchment properties and show no systematic trend across periods (Appendix C). We therefore opt for a framework in which A is modelled as a time-varying parameter while B, C' and D are treated as fixed constants.

Model. Mathematically, this Bayesian LAM with time-varying parameters is written as

$$C_i = A_{t_i} \cdot Q_i^{B-1} + C' \cdot Q_i^{D-1} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2) \tag{6}$$

Random walk structure. The trajectory of A_t is modelled on the log-scale to enforce positivity, defined as $\alpha_t = \log A_t$. This follows a first-order random walk structure [8] such that, for $t = 2, ..., N$:

$$\alpha_t = \alpha_{t-1} + \sigma_A \eta_t^A, \quad \eta_t^A \overset{\text{iid}}{\sim} \mathcal{N}(0, 1), \tag{7}$$

where σ_A is a variance hyperparameter controlling the rate of temporal change, such that larger values allow adaptation to changing catchment conditions.

Priors. Hyperparameters and initial α states are assigned the following priors:

$$\alpha_1 \sim \mathcal{N}(0, 2), \tag{8}$$

$$B \sim \mathcal{N}(0.5, 5)|_{[0,1]}, \quad D \sim \mathcal{N}(1.3, 0.5)|_{>1}, \tag{9}$$

$$\sigma \sim \mathcal{N}^+(0, 1), \quad \sigma_A \sim \mathcal{N}^+(0, 0.2). \tag{10}$$

The prior on α_1 was initialised to be non-informative across the range of plausible values of A from the baseline Bayesian LAM (evident by the range of A values, 0.27–15.33, for each site-determinand's first period) (Appendix C). Hyperprior σ_A is assigned weakly informative half-normal distributions, allowing plausible temporal drift in A_t. Priors on B and D are identical to the baseline LAM, following [2].

In the Bayesian LAM baseline model, the estimate of C' varies largely based on determinand, falling in the range of 0.05–0.11 for orthophosphate, and 1.76–3.90 for zinc. As C' varies substantially by determinand (0.05–0.11 for orthophosphate versus 1.76–3.90 for zinc) we use determinand-specific priors:

$$\text{Orthophosphate: } \log C' \sim \mathcal{N}(-2, 1) \tag{11}$$

$$\text{Zinc: } \log C' \sim \mathcal{N}(0.7, 1) \tag{12}$$

Model implementation. This model was constructed using `Stan` [18] and run in `R` [17].

5.1 Evaluation of Model

All four TV-LAM models achieved satisfactory convergence, with all scalar parameters obtaining $\hat{R} \leq 1.01$ and effective sample sizes of over 400 (with the exception of σ, obtaining 346 for the zinc model at the Thames) [11]. See Appendix F for convergence results of these scalar parameters for each model. No divergent transitions were observed.

Traceplots of these scalar parameters were investigated for each TV-LAM, with Orthophosphate at the Thames model shown in Figure 11 (with all other TV-LAM traceplots found in Appendix E.1).

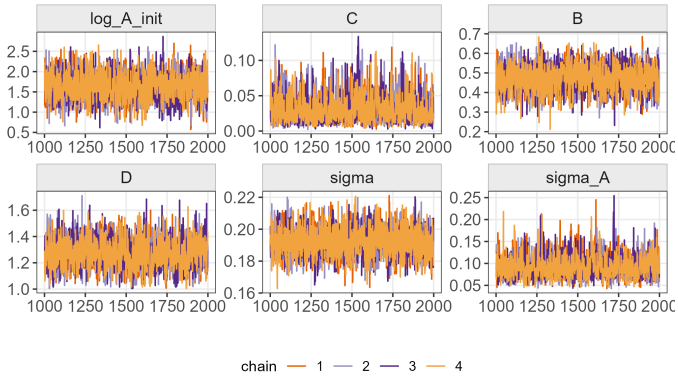


Figure 11: Traceplots of scalar parameters $\log A_{\text{init}}, B, C', D, \sigma, \sigma_A$ estimated for TV-LAM for orthophosphate at the Thames (constructed using Stan).

As shown in the traceplot, all scalar parameters experience good mixing, with each chain exhibiting stationary, well-mixed behaviour around a stable posterior region. Traceplots of other determinand-site TV-LAM combinations also showed good mixing, with B at both zinc sites exhibiting posterior mass concentrated near the upper boundary of its constrained support $[0, 1]$, indicating the constraint is active and that B is poorly identified for zinc (see Appendix E.1).

Residuals were plotted over time for each TV-LAM, with Orthophosphate at the Thames model shown in Figure 12 (with all other TV-LAM residual plots found in Appendix E.2).

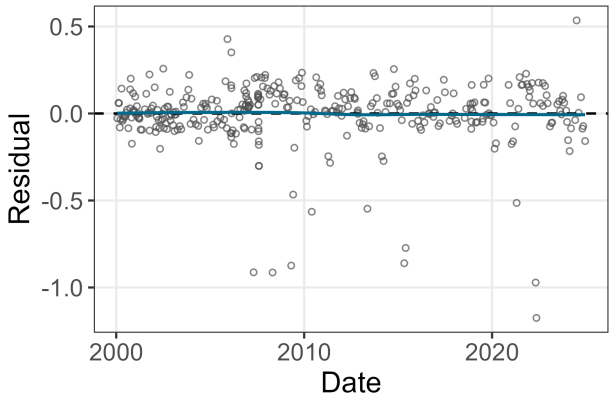


Figure 12: Bayesian TV-LAM Residual vs Time on Thames-Orthophosphate data with smoothed blue LOESS trend line.

The residual vs time plot of orthophosphate concentration at the Thames (Figure 12) is a key indicator that the time-varying LAM framework mitigates the prior issues caused by non-stationarity in our baseline Bayesian LAM (Figure 6). The LOESS smoothing line sticks consistently to the zero-line, indicating that our modelling framework mitigates any major systematic temporal trend. Other determinand-site combinations also exhibit adequate stationarity, with relatively flat LOESS lines across all site-determinand combinations (see Appendix E.2).

Posterior predictive checks are shown for orthophosphate at the Thames (Figure 13), with remaining site-determinand combinations in Appendix E.3.

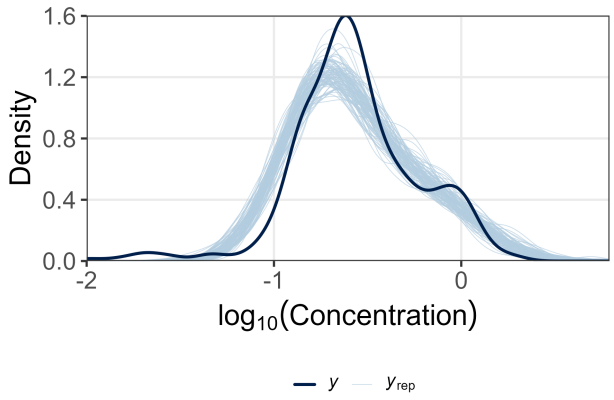


Figure 13: Posterior predictive density of simulated $\log_{10}$(Orthophosphate) concentration from TV-LAM at the Thames (light blue) and actual $\log_{10}$(Orthophosphate) concentration at the Thames (dark blue).

The posterior predictive density mostly envelopes the central mass of the observed Thames-orthophosphate concentration distribution but does not reproduce the bimodal structure, which likely reflects distinct regulation regime changes pooled across the full 25-year period. The Wyre-orthophosphate model also captured the observed density well. Both zinc models however produced simulated densities with peaks approximately half the height of the observed distribution, suggesting the models overestimate residual variance. This is likely due to a single fixed σ being unable to capture heteroskedasticity across a 25-year period with varying levels of change in A_t. Posterior predictive checks for the remaining site-determinand combinations (see Appendix E.3) show similar patterns, with orthophosphate fits capturing the observed density well and both zinc models overestimating residual variance.

5.2 Results

We now evaluate the point-source contribution to total pollutant load for each site-determinand TV-LAM. $\text{PS}_t\%$ is computed using Equation 5, evaluated at the full-period site-specific median flow Q_{med}, such that the resulting trajectory reflects only structural change in the point-source intercept A_t over time. This differs slightly from the windowed baseline LAM, where PS% is evaluated at the window-specific median flow, providing a more consistent reference condition

across the full 25-year period. Figures 14 and 15 present the posterior median trajectory of A_t with 80% credible intervals for orthophosphate and zinc respectively, providing the mechanistic basis for the PS% trends discussed below.

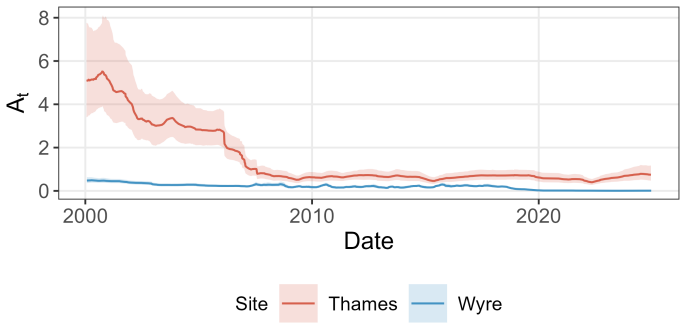


Figure 14: Posterior median estimates of A_t for orthophosphate at the Thames (red) and Wyre (blue) with 80% credible intervals.

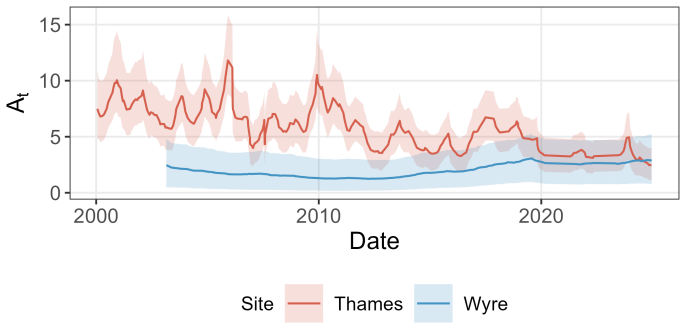


Figure 15: Posterior median estimates of A_t for zinc at the Thames (red) and Wyre (blue) with 80% credible intervals.

As the scale parameter of point-source inputs diminishes over time, point-source contribution to total load falls correspondingly, independently of changes in river flow. The Thames orthophosphate trajectory in particular, falling from approximately 5 in 2000 to below 1 by 2010 and stabilising thereafter, reflects a structural shift in point-source loading consistent with regulatory intervention rather than gradual ecological change.

Figures 16 and 17 present posterior median PS% estimates with 80% credible intervals for the Thames and Wyre sites.

Orthophosphate. The Thames shows clear point-source dominance from 2000–2006, with the 80% credible interval of the posterior estimate lying consistently above 75%. Point-source dominance then declines sharply between approximately 2007–2010, broadly coinciding with the mandatory introduction of phosphorus stripping at larger sewage treatment works under the Urban Wastewater Treatment Directive [16, 3], before stabilising at around 50% with 80% credible interval spanning 25–75% throughout to 2024.

The Wyre exhibits primarily diffuse-source dominance throughout the period, with initial posterior median estimates of point-source contribution between 25–55% and relatively narrow credible intervals of around 5-10% in width, reflecting the greater volume of observations at this site. From approximately 2018, there is a distinct and sustained

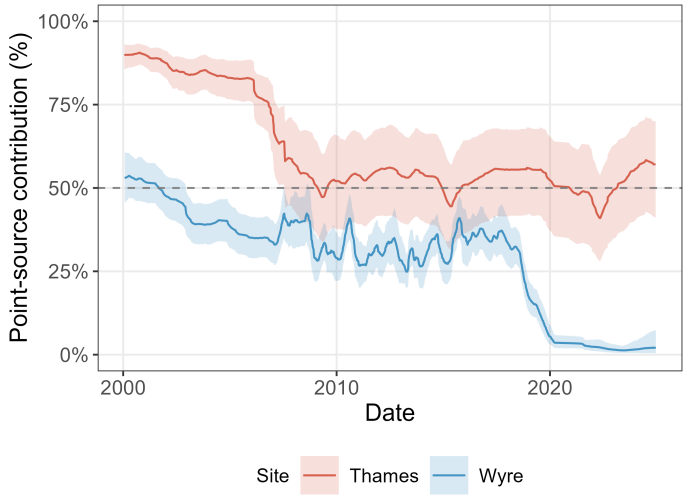


Figure 16: Posterior median estimates of point-source contribution (%) of orthophosphate in the Thames (red) and Wyre (blue). The corresponding shaded regions represent 80% credible interval in these posterior estimates.

decline in point-source contribution from around 30% to almost 0% by 2024, suggesting that total orthophosphate load in the Wyre at St Michaels has become almost entirely diffuse-source dominated by the end of the study period. This is consistent with its rural, agriculturally-dominated catchment.

These findings are broadly consistent with the windowed baseline LAM (Section 4.2), which estimated Thames orthophosphate point-source contribution declining from 72.8% (80% CI: 52.7–91.2%) in 2000–2004 to 25.2% (8.6–57.1%) by 2020–2024, suggesting suitable specification of the TV-LAM framework despite the methodological differences in flow reference condition.

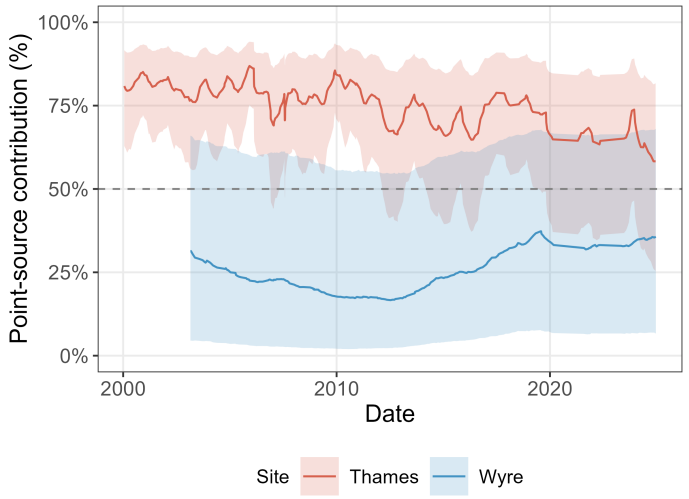


Figure 17: Posterior median estimates of point-source contribution (%) of zinc in the Thames (red) and Wyre (blue). The corresponding shaded regions represent 80% credible interval in these posterior estimates.

Zinc. Thames zinc remains predominantly point-source dominated throughout the period, with posterior median estimates broadly declining from around 75–80% in

2000 toward 50–60% by 2024, though 80% credible intervals remain wide, spanning approximately 50–90% in earlier periods and 30–85% by 2020–2024. The Wyre exhibits consistently lower point-source contribution, with posterior medians primarily in the region of 12–25% and very wide credible intervals spanning close to 0–60% for much of the period. The substantially wider credible intervals for both sites relative to orthophosphate reflect the larger proportion of censored zinc observations (around 20% at both sites) and the boundary behaviour of B identified in Section 5.1, which together limit the precision of source apportionment for this determinand. These zinc results should therefore be interpreted with considerably more caution than the orthophosphate findings.

These findings differ from the windowed baseline LAM (Section 4.2) in the estimated level of point-source contribution, with the baseline suggesting Thames zinc point-source contribution declined from 39.4% (80% CI: 18.7–62.7%) in 2000–2004 to 11.5% (4.8–25.5%) by 2020–2024, considerably lower than the TV-LAM trajectory. Both models nonetheless agree directionally, with point-source contribution declining over the period and uncertainty remaining high throughout.

6 Model Comparison

6.1 Residual Temporal Bias

As made evident in Section 4.1, a fundamental limitation of the windowed baseline Bayesian LAM is temporal non-stationarity. Even within five-year periods, a systematic downward LOESS trend in the residuals (Figure 6) indicates that the static parameter assumption is violated. By modelling A_t as a time-varying parameter following a first-order random walk, the TV-LAM explicitly accommodates this non-stationarity, eliminating the systematic residual bias as evidenced by the adequately flat LOESS smoothing line in Figure 12 (with corresponding plots for all other site-determinand combinations shown in Appendix E.2).

Beyond mitigating residual bias, the TV-LAM framework removes the need for arbitrary temporal windowing. In the baseline LAM, the stationarity assumption necessitates restricting model fitting to short time periods in which the flow-concentration relationship is approximately stable. The TV-LAM instead accommodates the full 25-year observation record within a single coherent model, eliminating the arbitrary choice of window boundaries and the information loss that occurs at each boundary transition.

6.2 In-Sample Goodness of Fit

In-sample goodness of fit for both modelling strategies is assessed using continuous ranked probability score (CRPS), a negatively-oriented scoring rule, computed on the 2000–2024 observation period [12]. CRPS evaluates a predicted probability distribution against the actual observed value, assessing both calibration and sharpness. This comparison is asymmetric: each baseline window is fitted only to the observations it is evaluated on, whereas the TV-LAM characterises the full record simultaneously, structurally favouring the baseline. The TV-LAM's CRPS advantage at the Thames sites is consequently a conservative lower bound on its true performance gain, while the baseline's marginal Wyre advantage may partly reflect this structural benefit rather than genuine superior predictive calibration.

Table 5 shows the mean CRPS values of the baseline LAM and TV-LAMs for each site-determinand combination.

Table 5: Mean CRPS comparison between the windowed baseline LAM and TV-LAM across all site-determinand combinations, evaluated on the shared 2000–2024 observation period. Lower values indicate better fit. Wyre zinc TV-LAM evaluated from 2005 onward to match baseline LAM time-windows. Bold denotes the better-fitting model.

Site	Determinand	Baseline	TV-LAM
Thames	Orthophosphate	0.105	**0.089**
Wyre	Orthophosphate	**0.089**	0.095
Thames	Zinc	0.079	**0.065**
Wyre	Zinc	**0.076**	0.082

The TV-LAM achieves lower mean CRPS for both Thames orthophosphate and Thames zinc, the two site-determinand combinations exhibiting the strongest long-run decline in A_t (Figures 14 and 15). The baseline achieves lower CRPS for Wyre orthophosphate and zinc, where A_t remains comparatively stable throughout the period, and thus the windowed approach is well-specified within each five-year period. These results are ecologically coherent: the TV-LAM's advantage materialises precisely where structural non-stationarity is greatest, and is absent where the baseline's simpler parameterisation is adequate.

Notably, the baseline's CRPS advantage at the Wyre is marginal: 0.089 vs 0.095 (reduction of approximately 6.3%) for orthophosphate and 0.076 vs 0.082 (7.9%) for zinc, whereas the TV-LAM's advantage at the Thames is more pronounced, with CRPS reductions of 15.2% for orthophosphate and 17.7% for zinc relative to the baseline. Given that each baseline window is fitted to approximately 20% of the observations available to the TV-LAM, these CRPS reductions represent a meaningful improvement in goodness of fit by a model operating under greater constraint.

Figure 18 shows observation-level CRPS for the windowed baseline LAM and TV-LAMs for orthophosphate at the Thames.

Figure 18 reveals that the aggregate CRPS differences in Table 5 are not uniformly distributed across the observation period but are concentrated in specific temporal regions. For Thames orthophosphate, the TV-LAM's advantage is most pronounced between approximately 2007–2014, a period of rapid decline in A_t (Figure 14), and where the baseline model is most severely stressed by non-stationarity. For Thames zinc, the TV-LAM maintains a consistent advantage throughout the period, consistent with the persistent but noisier downward trend in A_t at this site. For the Wyre site-determinand combinations, the LOESS lines cross repeatedly with no sustained advantage for either model, corroborating the marginal aggregate differences reported

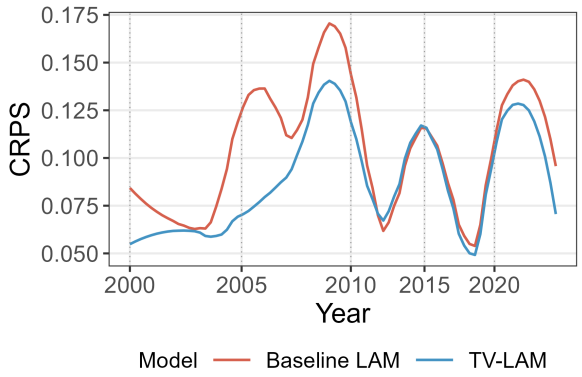


Figure 18: LOESS-smoothed observation-level CRPS for the windowed baseline LAM (red) and TV-LAM (blue) for orthophosphate at the Thames.

in Table 5. Taken together, Figures 14 and 18 establish that the TV-LAM's goodness-of-fit advantage is mechanistically linked to the rate of structural change in point-source inputs: where A_t evolves rapidly, the TV-LAM characterises the concentration record more accurately than a sequence of static windowed models.

Overall, these results indicate that the TV-LAM is the preferred framework for characterising source apportionment over non-stationary long-term periods, while the windowed baseline remains adequate where catchment conditions are approximately stable. Whether these patterns generalise beyond the Thames and Wyre, and what they imply for EA monitoring and investment decisions, is considered in the Discussion (Section 7).

7 Discussion

7.1 Main Findings

The central finding of this analysis is that the Bayesian LAM and TV-LAM frameworks together document ecologically coherent, site and determinand-specific trajectories in point-source contribution over the 2000–2024 period.

For orthophosphate, both frameworks document a regulatory success story consistent with the progressive implementation of phosphorus stripping at larger STWs under the Urban Wastewater Treatment Directive [16, 3]. At the Wyre, TV-LAM posterior estimates shift from approximately 50% point-source contribution in 2000 to near 0% by 2024, reflecting near-complete elimination of STW dominance in a predominantly rural catchment with comparatively few and smaller STW discharges. At the Thames, point-source contribution declines sharply between 2007—2010 before stabilising at around 50%, suggesting that the substantially larger aggregate STW load on this major urban catchment prevents diffuse-source dominance from being achieved, an outcome broadly consistent with national HMS trends and prior LAM studies on UK lowland rivers [2, 4].

For zinc, both models indicate persistent point-source dominance at the Thames and diffuse-source dominance at the Wyre throughout, though wide 80% credible intervals limit the precision of these apportionments considerably. While the directional agreement between models provides some confidence, the credible interval width is such that compliance-relevant thresholds cannot be reliably distinguished. From an EA operational perspective, the wide credible intervals mean point and diffuse contributions to zinc cannot be distinguished with sufficient precision for compliance decisions at either site.

The TV-LAM successfully characterises these trajectories across the full 25-year study period, and eliminates the systematic residual temporal bias that persists even within five-year windows in the baseline Bayesian LAM. TV-LAM consistently fitting our data better than baseline LAM in periods of large structural change in A_t indicates that the random walk prior successfully captures non-stationarity rather than absorbing residual variation. Where catchment structure is stable, as at the Wyre, the simpler windowed baseline remains an adequate and more parsimonious alternative.

7.2 Limitations

Several limitations of this analysis warrant acknowledgement. The TV-LAM treats C' as fixed because modelling it as time-varying alongside A_t produced posterior non-identifiability, indicating the data cannot simultaneously identify two freely-evolving trajectories from a single concentration-flow relationship. Consequently, the framework cannot detect structural changes in diffuse-source inputs over time, a meaningful ecological caveat particularly for the Wyre, where agricultural land management practices have evolved considerably over the study period.

For zinc specifically, B was found to concentrate posterior mass at the upper boundary of its constrained support $[0, 1]$ at both sites, indicating the point/diffuse decomposition may itself be poorly identified for this determinand. This likely reflects a structural limitation of applying the LAM to zinc: B concentrating at its upper boundary suggests the point-source term is becoming load-increasing with flow, which is physically inconsistent with dilution of a constant point-source input and indicates the two-term power-law decomposition may be poorly suited to this determinand. The wide credible intervals for zinc should therefore be understood as reflecting both high censoring rates and the weaker hydrological separability of point and diffuse sources for zinc.

Furthermore, posterior predictive checks revealed that all TV-LAMs overestimated residual variance, likely because a single fixed σ cannot capture heteroskedasticity across a 25-year period of varying structural change in A_t. CRPS values should also be interpreted as comparative goodness-of-fit metrics on training data only: each baseline window is fitted to approximately 20% of the observations the TV-LAM must simultaneously characterise, conferring a structural training advantage on the baseline that precludes direct comparison as a measure of predictive ability. Inferring out-of-sample performance would require leave-one-out cross-validation on matched held-out data. Results span only two sites and two determinands, and as such findings should not be directly

extrapolated to other catchments.

7.3 Future Directions

The TV-LAM framework developed here was applied to two sites and two determinands; extending this to all 135 HMS sites across England would establish whether the long-term declines in point-source contribution identified at the Thames and Wyre are representative of the broader river network. This would allow the EA to identify catchments that remain point-source dominated for orthophosphate, and thus inform decisions on where further STW investment would yield the greatest water quality improvement. From a modelling perspective, the fixed σ assumption could be addressed by treating observational variance as time-varying, better capturing heteroskedasticity across periods of rapid structural change in A_t. For zinc specifically, future work might consider whether alternative model formulations would yield better-identified source apportionment. The modified LAM of [7], which accounts for upstream inflow contributions, represents one such candidate for determinands with more complex catchment source pathways.

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